

Question Number	Answer	Additional guidance	Mark
6(a)	An explanation that includes four of the following points: • water for {hydrolysis reactions / form cytoplasm / solvent / hydration} (1) • (some bacteria require) oxygen for (aerobic) respiration (1) • (suitable/optimum) temperature for {enzyme-controlled / metabolic} reactions to occur (at suitable rate) (1) • (suitable/optimum) pH for {enzyme-controlled / metabolic} reactions to occur (at suitable rate) (1) • named organic molecule and an explanation of why that organic molecule is needed (1)	ACCEPT (some anaerobic bacteria require) no oxygen for anaerobic respiration ACCEPT (suitable) temperature so enzymes do not denature ACCEPT (suitable) pH so enzymes do not denature e.g. glucose for respiration / lipids to provide fatty acid chains for new cell membrane / proteins to provide amino acids for {protein synthesis / growth}	(4)

Question Number	Answer	Additional guidance	Mark
6(b)(i)	A calculation showing the following steps: • calculation of difference in mean area of ulcer (1) • calculation of percentage decrease (1)	<u>Example of calculation</u> $(802.71 - 95.71) = 707$ $\frac{(802.71 - 95.71)}{802.71} \times 100 = 88 / 88.077 / 88.08 / 88.1 (\%)$ Correct answer with no working shown scores full marks	(2)

Question Number	Answer	
6 (b)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. <u>Description of data in table</u> • mass of mucus • ulcer area <u>Conclusions and Explanations</u> • omeprazole was most effective treatment /none of the extracts were as effective as omeprazole at reducing ulcer area / all treatments resulted in {reduction of ulcer area / increased mass of mucus} • correlation between mucus mass and ulcer area (apart from omeprazole) • bacteria damage stomach lining causing ulcers • suggestion that {omeprazole / extract} increases mucus production • high mucus content prevents bacteria {reaching / damaging} stomach lining • explanation of how mucus production links to ulcer area • high mucus content prevents acid contents damaging stomach lining <u>Sustained</u> • link to biological process resulting in increased mucus production • {omeprazole / extract} causes the {stomach lining / goblet cells} increases mucus production • suggestion that {omeprazole / extract} reduces acidity of stomach making it less suitable for <i>H. pylori</i> • antimicrobial properties of {omeprazole / extract} reducing {number / reproduction} of bacteria • therefore {slow population growth / reduced population of bacteria} results in {reduced damage to stomach lining / smaller ulcer area}	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Either Description of: mass of mucus data and/or ulcer area data OR description of one and/or conclusion / explanation of one data set
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	description of both data plus: explanation of mass of mucus / ulcer area and/or simple conclusion of most effective treatment
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	all level 2 plus: a detailed explanation using sustained application. The detail of sustained application determines the mark in this level.